HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Mapped **150+ forestry plots** across various land parcels using ArcGIS, ensuring accurate spatial representation for management planning.
- Processed and analyzed **LiDAR data** to identify tree canopy height variations and potential growth areas for over 50 key forest sites.
- Managed and updated **geodatabases** for forestry operations, incorporating new survey data and maintaining spatial integrity for **ArcGIS**.
- Supported the collection and analysis of geographical data, including **LiDAR**, for wind and solar project assessments, contributing to risk identification.
- Prepared detailed **forestry maps** and supporting documentation for submission to forest management authorities, adhering to specific compliance standards.
- Maintained accurate **GIS records** and assisted in the management of **GPS equipment** and associated databases for field teams.
- Developed **Python scripts** to automate the georeferencing of incoming survey data, reducing manual processing time by approximately **30%**.

SKILLS

GIS Software – ArcGIS, QGIS, ArcGIS Pro, ArcGIS Online

Spatial Data Analysis – LiDAR, Geodatabases, Spatial Databases, Coordinate Systems, Georeferencing

Programming & Scripting – Python, SQL, Bash Scripting

Data Management – Database Maintenance, Data Collection, GPS Equipment Management

Mapping & Visualization – Forestry Maps, Geospatial Data Presentation, Map Creation

Cloud Platforms – AWS (EC2, S3, CloudWatch)

PROJECTS

Forestry Land Management System Enhancement

- Developed Python scripts to automate the processing of survey data into the central **geodatabase**, improving data ingestion efficiency by **20%**.
- Designed and implemented map outputs in **ArcGIS** for forest managers, highlighting areas for silvicultural intervention based on spatial analysis.
- Managed the lifecycle of spatial datasets, ensuring consistency and accuracy across different versions of the forestry land database.

LiDAR Data Processing for Tree Height Analysis

- Engineered a workflow to process raw **LiDAR** point clouds, extracting canopy height models for a **100-hectare** forestry concession.
- Performed spatial analysis in **Python** to identify and segment individual tree crowns within processed LiDAR data, supporting growth assessments.
- Created visualisations and summary reports of tree height distributions for use by senior foresters and land managers.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)